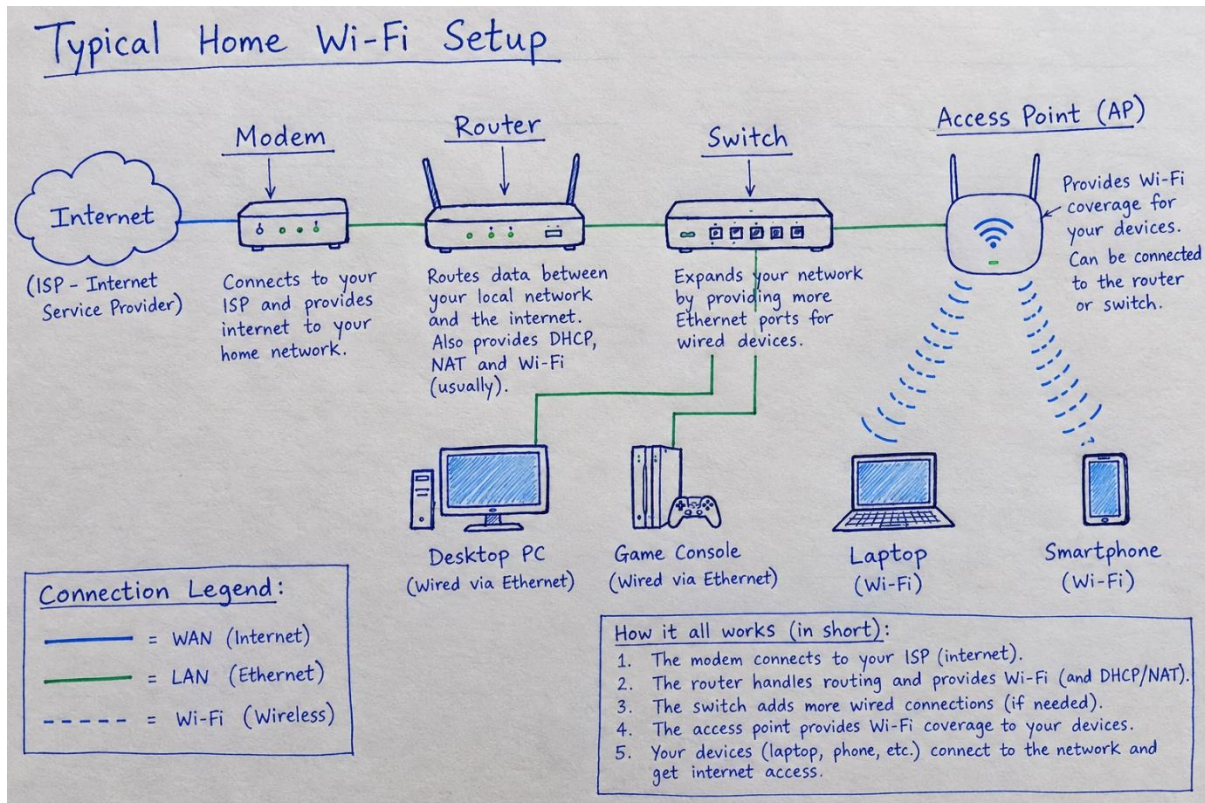


Day 1 : Introduction to Networking Hardware Theory

Task 1 : Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



- A typical home Wi-Fi network uses a **modem, router, switch, and access point** to connect different devices.
- The modem connects the home network to the ISP and provides Internet access.
- The router manages network traffic and connects the local network to the Internet.
- A switch provides additional Ethernet ports for wired devices such as a desktop PC or game console.
- The access point provides Wi-Fi connectivity for wireless devices such as a laptop and smartphone.

Network flow:

ISP → Modem → Router → Switch → Wired Devices

Router → Access Point → Laptop / Smartphone

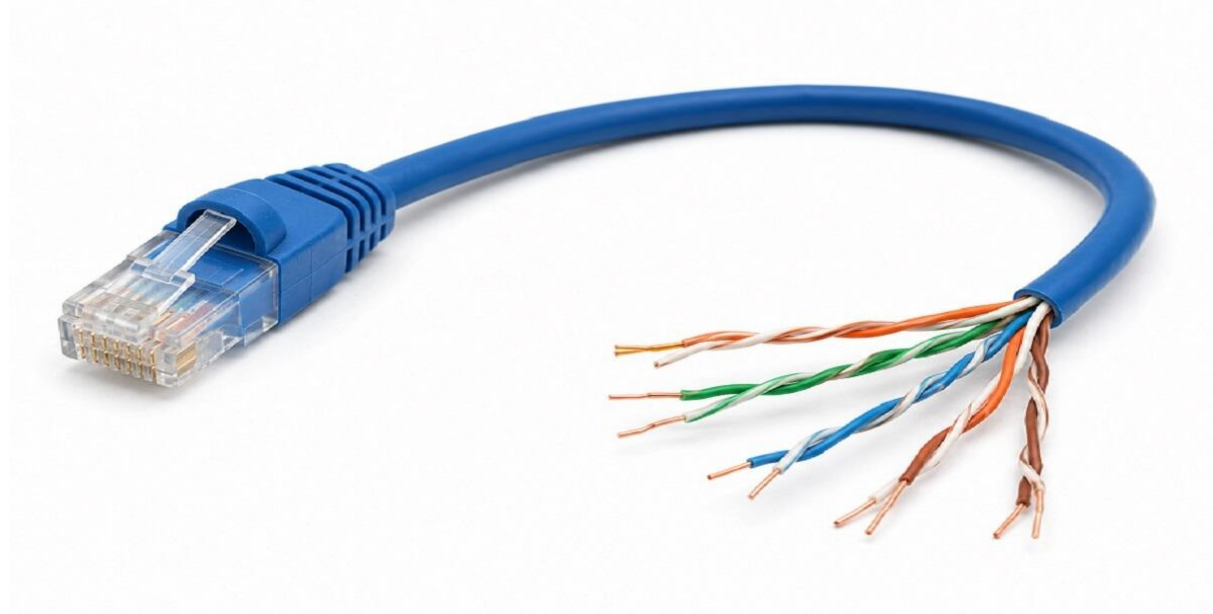
This setup allows both wired and wireless devices to communicate with each other and access the Internet

Task 2 : List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Device	Main Function	Real-Life Example
Modem	Acts as the gateway between the ISP and your home network . It receives the Internet signal from the service provider.	At home: A fiber/cable modem receives the Internet connection from the ISP.
Router	Works like a traffic manager . It decides where data should go and allows multiple devices to share one Internet connection.	At home: A Wi-Fi router lets a phone, laptop, and Smart TV use the same Internet connection.
Switch	Works like a connection hub for wired devices , sending network data to the required device instead of broadcasting it everywhere.	In a computer lab: One switch can connect many desktop PCs to the same LAN.
Access Point	Creates a wireless entry point into the network and helps provide Wi-Fi coverage in areas where the router's signal is weak.	In a college/office: Access points provide Wi-Fi in classrooms, corridors, or different floors.
Network Cable	Provides a physical path for data between networking devices, giving a stable wired connection.	For gaming: An Ethernet cable connects a gaming PC directly to a router or switch for a stable connection.

Task 3 : Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

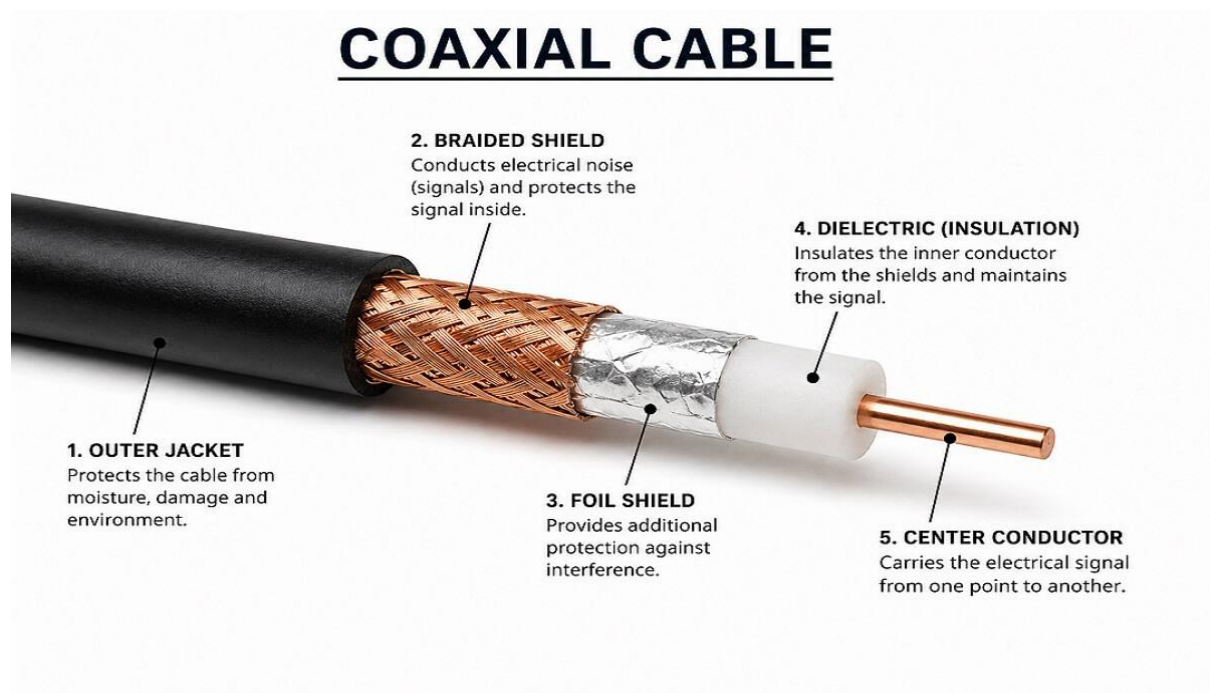
1. Ethernet Cable (Twisted Pair)



Common Use: Used to connect **computers, routers, switches, and other devices in a LAN**, such as in homes, offices, and computer labs.

Source : <https://www.knowcomputing.com/advantages-disadvantages-characteristics-twisted-pair-cable/>

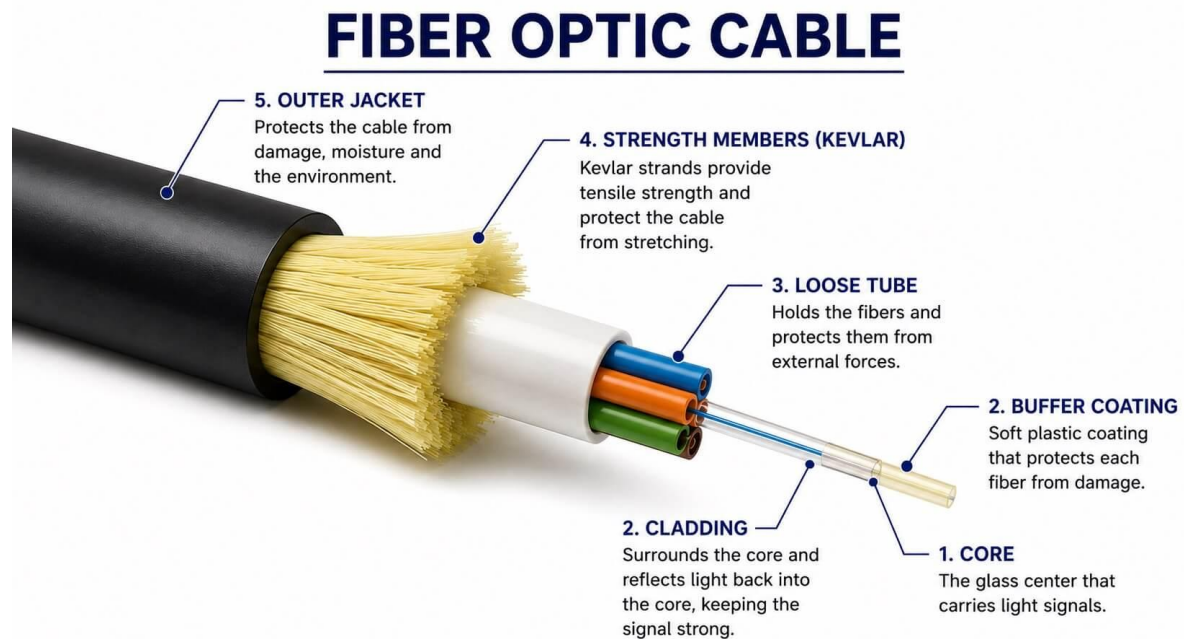
2. Coaxial Cable



Common Use: Commonly used for **cable TV** and **broadband Internet connections**, including connections to cable modems.

Source <https://www.knowcomputing.com/coaxial-cable-uses-parts-advantages/>

3. Fiber Optic Cable



Common Use: Used for **high-speed and long-distance Internet communication**, such as telecommunications, data centers, and Internet backbone networks.

Source: <https://www.knowcomputing.com/fiber-optic-cable/>

Task 4 : Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Basis	Router	Switch
1. Main Use	Connects different networks and provides communication between them.	Connects multiple devices within the same network (LAN) .
2. Typical Location	Commonly found at homes, offices, and Internet gateways .	Commonly found in computer labs, offices, and data centers .
3. Data Handling	Uses IP addresses to decide where to forward data packets.	Uses MAC addresses to forward data frames to the correct device.
4. Network Layer	Mainly works at Layer 3 (Network Layer) of the OSI model.	Mainly works at Layer 2 (Data Link Layer) of the OSI model.
5. Example	A home router connects the home LAN to the Internet .	A switch connects PCs, printers, and other wired devices in an office LAN.
6. Traffic Management	Routes traffic between different networks .	Forwards traffic between devices on the same LAN .